

BANGLADESH UNIVERSITY OF ENGINEERING AND
TECHNOLOGY (BUET)

Department of Computer Science and Engineering

Comparative Analysis of Neural Network Architectures

CSE 472: Machine Learning Sessional

Submitted By:

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1 Neural Network Architectures

Four distinct Feedforward Neural Network (FNN) architectures were designed to explore the impact of depth, width, and regularization on the medical students dataset. All models utilize the **ReLU (Rectified Linear Unit)** activation function for hidden layers and **Cross-Entropy Loss** for optimization.

- **FNN_2Layer (Base Model):** A shallow architecture consisting of 2 hidden layers.
 - Layer 1: 64 neurons, ReLU
 - Layer 2: 32 neurons, ReLU
- **FNN_3Layer (Deep Model):** A deeper network designed to capture more complex feature interactions.
 - Layer 1: 128 neurons, ReLU
 - Layer 2: 64 neurons, ReLU
 - Layer 3: 32 neurons, ReLU
- **FNN_Dropout (Regularized Model):** Based on a 2-hidden layer structure but incorporates Dropout layers to prevent overfitting.
 - Layer 1: 128 neurons, ReLU, Dropout ($p = 0.3$)
 - Layer 2: 64 neurons, ReLU, Dropout ($p = 0.3$)
- **FNN_Wide (Wide Model):** A high-capacity shallow model with a significantly larger first hidden layer.
 - Layer 1: 256 neurons, ReLU
 - Layer 2: 64 neurons, ReLU

2 Experimental Results and Visualizations

The models were trained using both *StandardScaler* and *MinMaxScaler*. Below are the loss curves for the top 4 performing configurations based on validation accuracy.

2.1 Top 4 Performing Models: Training and Validation Loss

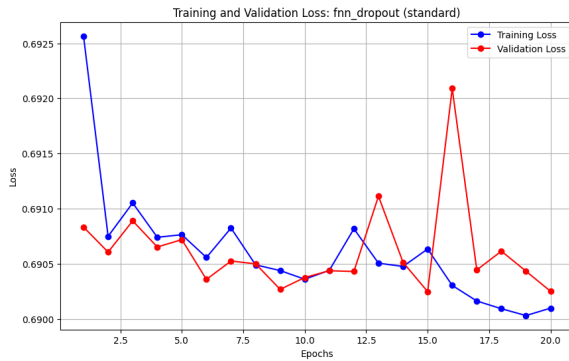


Figure 1: FNN_Dropout (Standard)

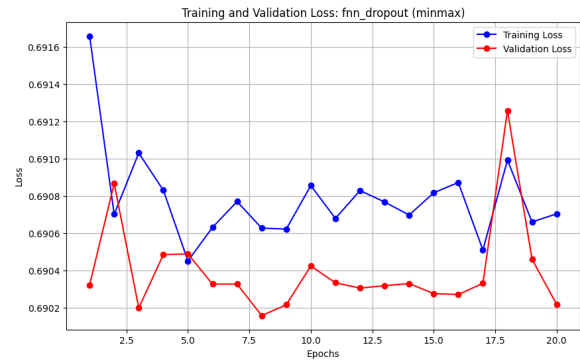


Figure 2: FNN_Dropout (MinMax)

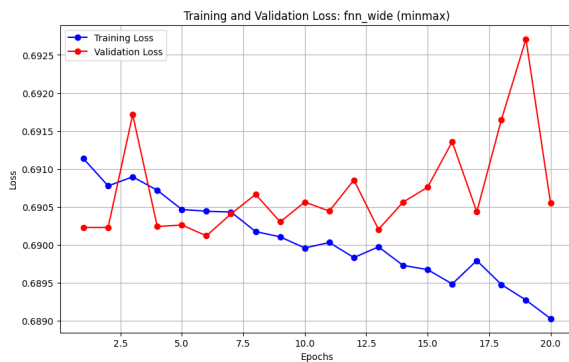


Figure 3: FNN_Wide (MinMax)

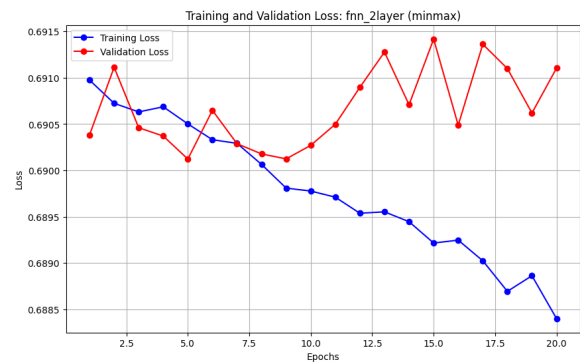


Figure 4: FNN_2Layer (MinMax)

3 Justification of Selected Architecture

The **FNN_Dropout** model (using either Standard or MinMax scaling) was selected as the final architecture due to its superior validation accuracy of approximately **89.97%**.

3.1 Performance Discussion

Despite the high accuracy, the final test set performance reveals a critical issue:

- **Accuracy:** 0.8997
- **Precision/F1-score:** 0.0000
- **AUROC:** 0.5030

The discrepancy between high accuracy and zero F1-score suggests a severe **class imbalance** in the dataset. The model has likely defaulted to predicting the majority class for every instance. While *FNN_Dropout* was the most robust architecture against overfitting (as seen in its loss curves), the AUROC of ~ 0.5 indicates the model is currently performing no better than random guessing for the minority class.

3.2 Final ROC Curve

The ROC curve further illustrates the model's struggle to distinguish between classes, maintaining a nearly diagonal trajectory.

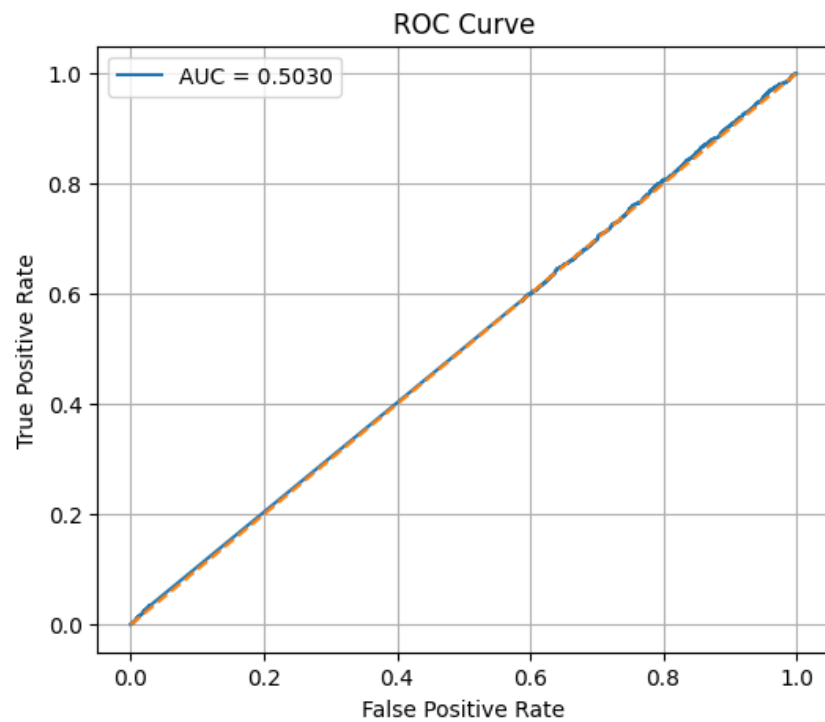


Figure 5: ROC Curve for the Final Selected Model